

Assembly language → low level programming language for computer

* Assembly " is specific a particular computer Architecture

high level language Portable across Multiple Architecture require compiling or interpreting

Assembly language is converted into executable Machine code by Assembler

Source Program → Assembly language Program

Object " → output of Assembler

Executable " → output of linker

Assembling

First Translating Source code into Object Code OBJ File

Assembler → calculates offset of every Data in DS
↳ and every instruction of CS

Create header

Part of header contain Infor about
Incomplete Addresses

EXE	Com
size big	Small 64 K or less
	one segment both data and instruction
	PSP, stack
	Start of PSP
	ORG 100H
	one segment both data and instruction

Segments: Special Area defined in Memory begin on Paragraph boundary

Main Segments

- ① Code Segment (CS) → Machine Instruction
- ② Data " (DS) → Program Data and Constants
- ③ Stack " (SS) → contains Data or Address needed by called Subroutine (Method)

CS, DS, SS Contain Addesses of Segments

Actual Address = Segment Address + offset
 Physical " = beginning of Address + actual data Address

General registers

Segment " CS, DS, SS, ES, FS, and GS

Pointer " →

Address of next Instruction → address incs + offset in PC
 Instruction Pointer (IP)

Address of current word in stack = address SS + offset
 address in SP

Stack Pointer

General Purpose register

AX (Accumulator): I/O and Arithmetic

BX (base reg) : index to extended Address

CX (Counter) : For looping

DX (Data reg) : for I/O operation

Index register

SI Source Index

DI Destination II

Flag registers

OF → Overflow

DF → direction For moving or comparing string

IF → Interrupt

SF → Sign (0 → +, 1 → -)

ZF → result of Arithmetic or comparison
(0 = non zero, 1 = zero)

CF → Carry

PF → Parity (odd or even)

AF → Auxiliary carry contain carry out of 3rd or 8th bit data

TF → Trap Set single instruction

Comment → Semicolon (;)

Reserved word also

MOV, ADD, END, SEGMENT FAR
Size (6) Data, (6) Model
PAGE

Name → first character must be Alphabetic
 or special character expect period
 must not start with number. see

Name refer to Address, Data Item

label " " " Instruction

Operation → operand (if any) → Information for the
 operation to act on → For Data Item, Initial
 value
 for instruction → where to perform action

* Ret → NO operands

one operands → MUL 10

2 " → MOV CX, 10

* (3) operands → SHRD ECX, EBX, CL

Data Item → Undefined Value ?
 → Constant
 → Character String
 → Numeric Value

Name Dn Expression

Dn → Directive
 DB byte DW → word
 DD Double word

Expression → uninitialized ?
 → Constant

name Dn repeat count DUP (expression)
 مكرر

FILL DW 10 DUP (55)

55 55 55 55 55 55 55 55 55

Statement → 2 types of Statement

Instruction MOV, ADD

assembler translates to object code

Directive → assembler to perform specific Action

Directive → COUNT DB 1

Instruction → MOV AX, 0

Directive Control the way Source Program assemble list
NO Machine Code

6 Directive →
→ Page
→ Title
→ Segment
→ procedure
→ Assume
→ END

Segment Directive

• EXE Format

Stack Segment	define	stack storage
Data	//	Data items
Code	//	Provides for executable code

(Name) Segment [Options]

// ENDS

① Alignment type: Indicate boundary
 (*) type on which segment to Begin
 PARA Used default

② Combine type: Combine the Segment
 with other segments
 linked after Assembly
 type Stack, Public, AT, Common

Public and Common → Separated assembly
 Program when linking

program is NOT to be linked NOVE

③ Class type

Group related segment when linking

Code Port $\rightarrow$ CS
 data " $\rightarrow$ DS
 Stack " $\rightarrow$ SS

Procedure

ProcedureName PROC Operand

" " ENDP

Operand $\rightarrow$ relates to Program execution
EAR

Assume directive

Tells the Assembler the Purpose of each segment in program

ASSUME SS: ~~Stack~~^{Stack}, DS: ~~data~~^{data}, CS: ~~code~~^{code}
 Comma

END Directive

ENDS $\rightarrow$ end segment

ENDP $\rightarrow$ end procedure

END $\rightarrow$ End of program

DEC AX

نشرح واره
subtract 1 from single AX operand

INC AX Ax واره

DEC, INC Single operand

MOV, AX, 4C00H

INT 21H

End Processing

صل على النبي

لا تنسوا أبي وجدتي من دعائكم

لا تنسوني من صالح دعائكم